Master's Thesis · Industrial Informatics · SS 2026

Generating User-Relevant Queries from Detected Amenities in Accommodation Images

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Agenda

Introduction

Generating User-Relevant Queries from Detected Amenities in Accommodation Images

Accommodation Images + **Detected Amenities** + **Generating User-Relevant Queries**

↓ ↓ ↓

Images | Detection | Queries

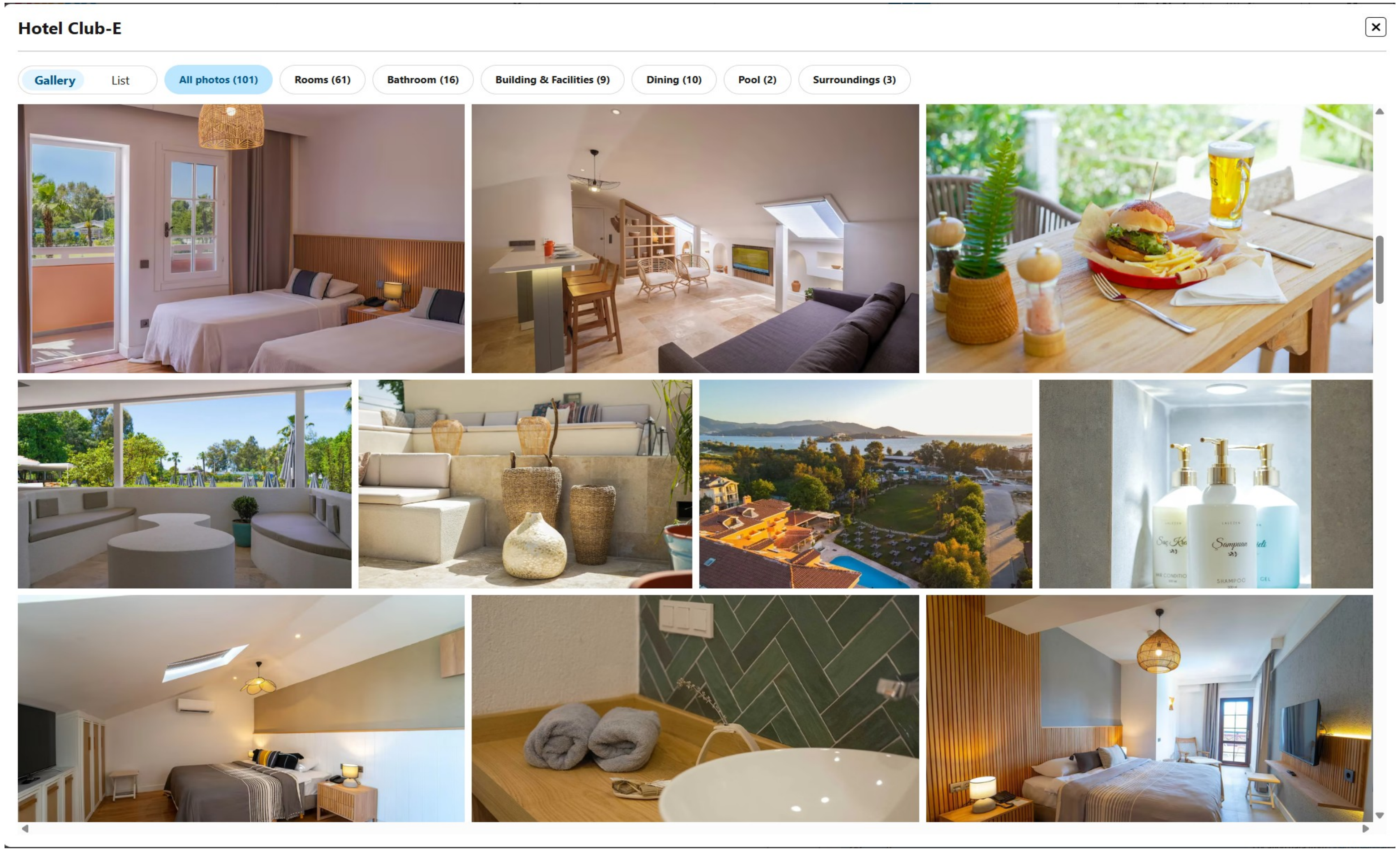
Problem & Motivation

The Missing Layer

Current platforms show images. No system tells users what to look for, or what to ask.

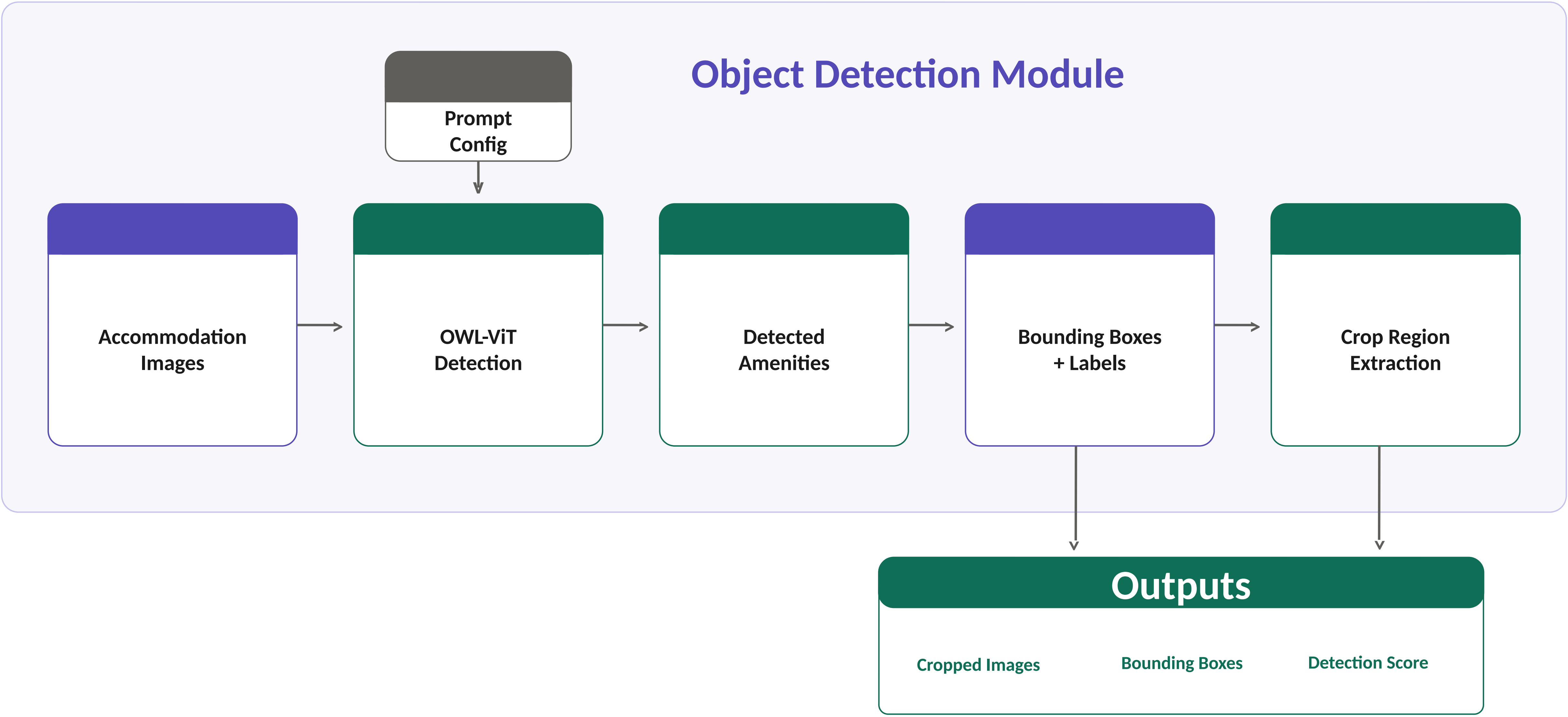
The Visual Reality

- **Images drive decisions**
- **Volume is overwhelming**
- **Amenities stay hidden**
- **Every traveller is different**
- **No personalised guidance**

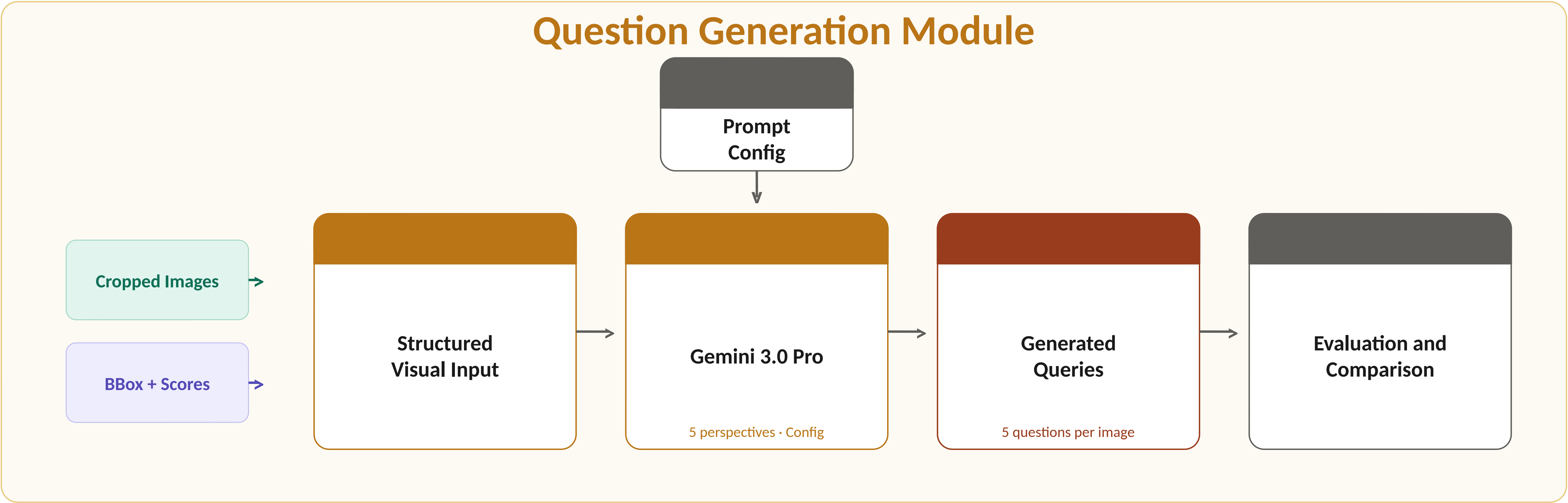


Source: trivago accommodation listing, accessed May 2026.
https://www.trivago.com/

System Pipeline Overview



System Pipeline Overview



Data & Amenity Selection

Data Source

trivago internal dataset

Real accommodation images from well-curated properties, high quality and commercially relevant

Unseen by all models

No model was trained or fine-tuned on this data, **results reflect genuine generalisation**

Negative images included

Each amenity set contains images where the target object is absent, testing false positive behaviour

Amenity	Number of Images	True Positive	False Positive
Bathtub	292	216	76
Kettle	273	201	72
TV	338	247	91
Hairdryer	290	215	75
Mirror	378	285	93

Data & Amenity Selection

Amenity Selection

Three selection criteria

1. Dataset feasibility
2. Visual detectability
3. User relevance

Five final amenities

Kettle · Bathtub · TV · Hairdryer · Mirror

Annotation via Roboflow

Ground-truth bounding boxes created manually via annotation

Key Principle

Every decision, from data selection to amenity choice, was made to reflect real-world search conditions.

OWL-ViT & Prompt Design

OWL-ViT · Vision Transformer · Open-vocabulary · Zero-shot · No retraining required

Prompt Configurations

Baseline

Amenity name only

bathtub · kettle · mirror

Descriptive

Name + context

electric kettle with a power cord in a hotel room

Variants

Multiple phrases

hairdryer · wall-mounted hairdryer

Threshold & Pipeline Role

0.05 → 0.45

Confidence sweep per amenity per prompt

IoU threshold: 0.50

Detection Gates Generation

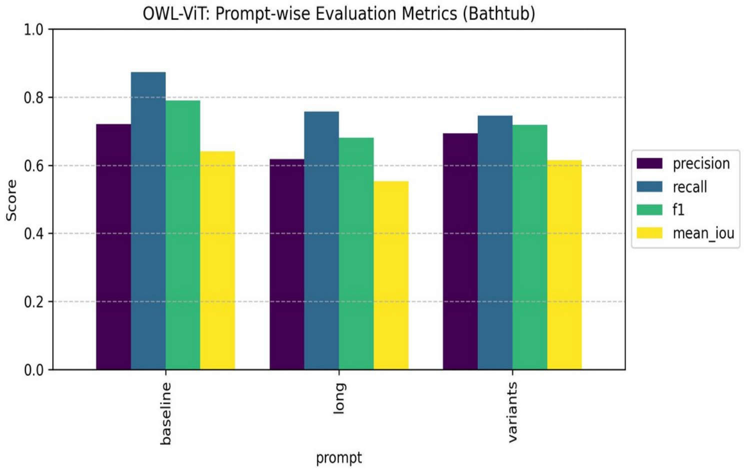
Only detected amenities proceed

False positive: irrelevant question generated

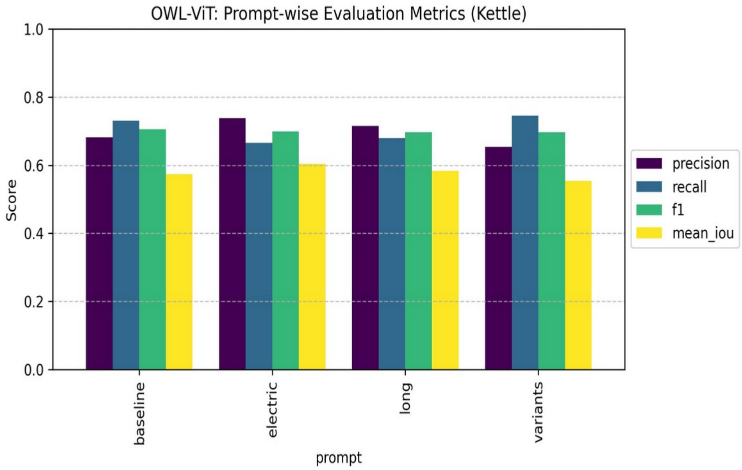
Detection Results · Prompt Comparison Across Amenities

Each chart shows Precision · Recall · F1 · Mean IoU across prompt configurations at optimal threshold

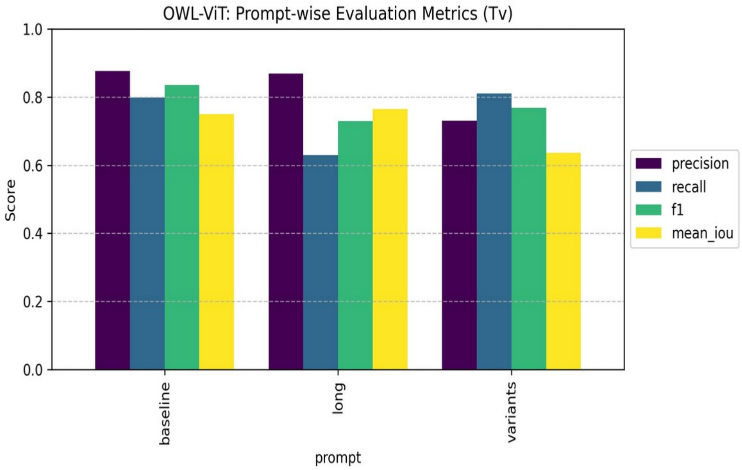
Bathtub



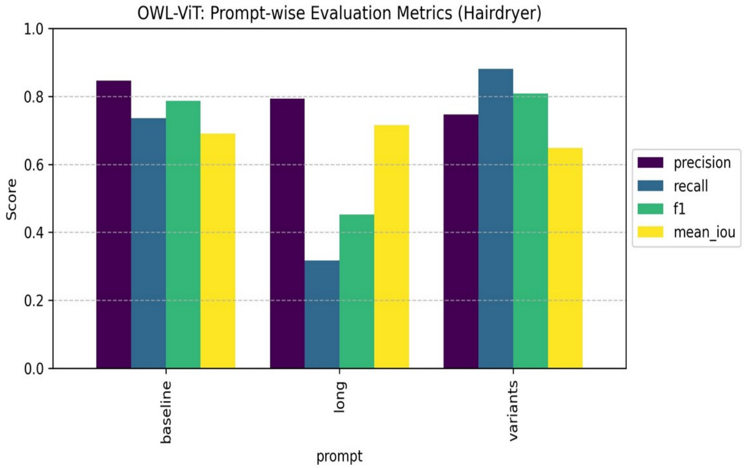
Kettle



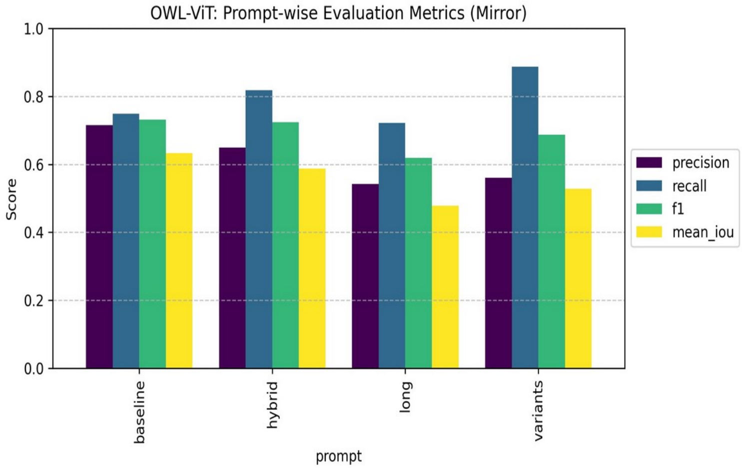
TV



Hairdryer



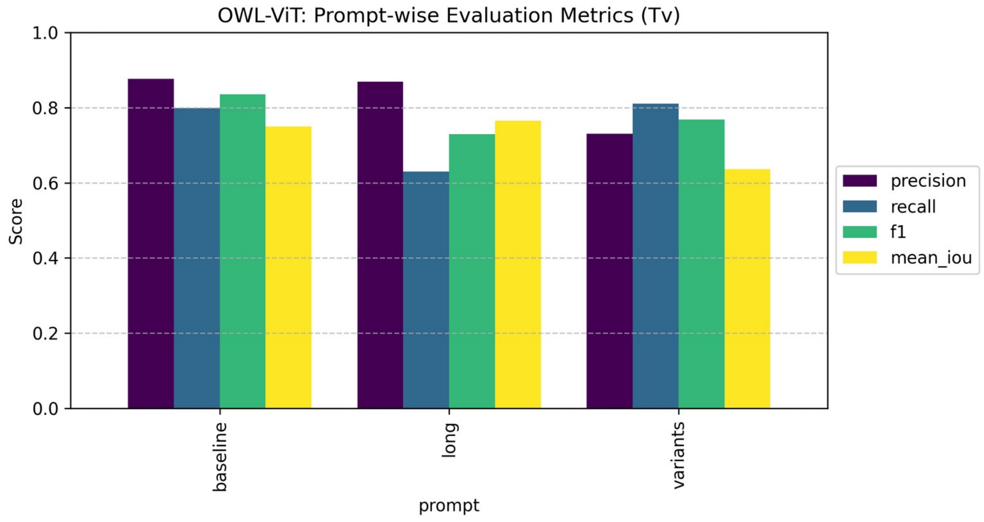
Mirror



Detection Results · Key Takeaways

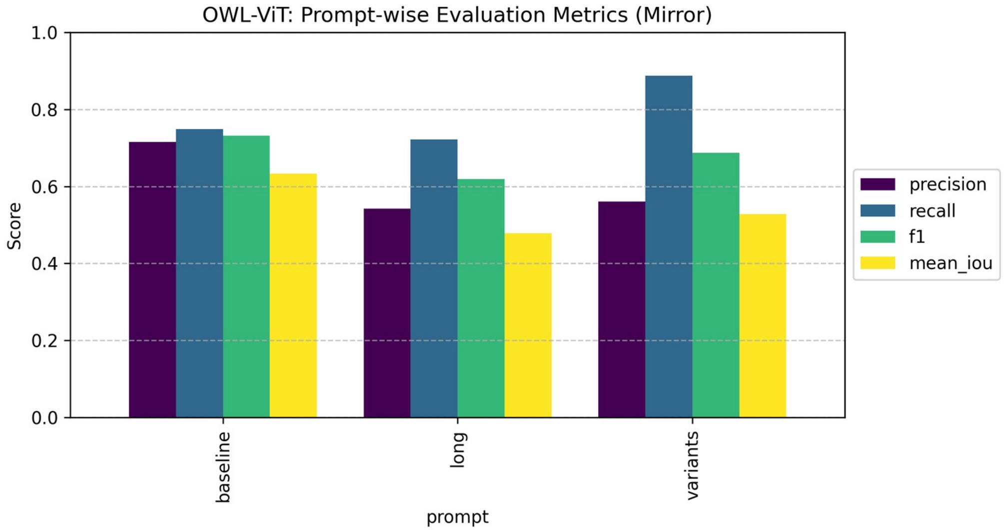
Best case · visually standardised

Baseline · IoU **0.750**
best localization



Hardest case · reflective · visually ambiguous · high false positive rate

Baseline · IoU **0.633**
best localization despite lower recall



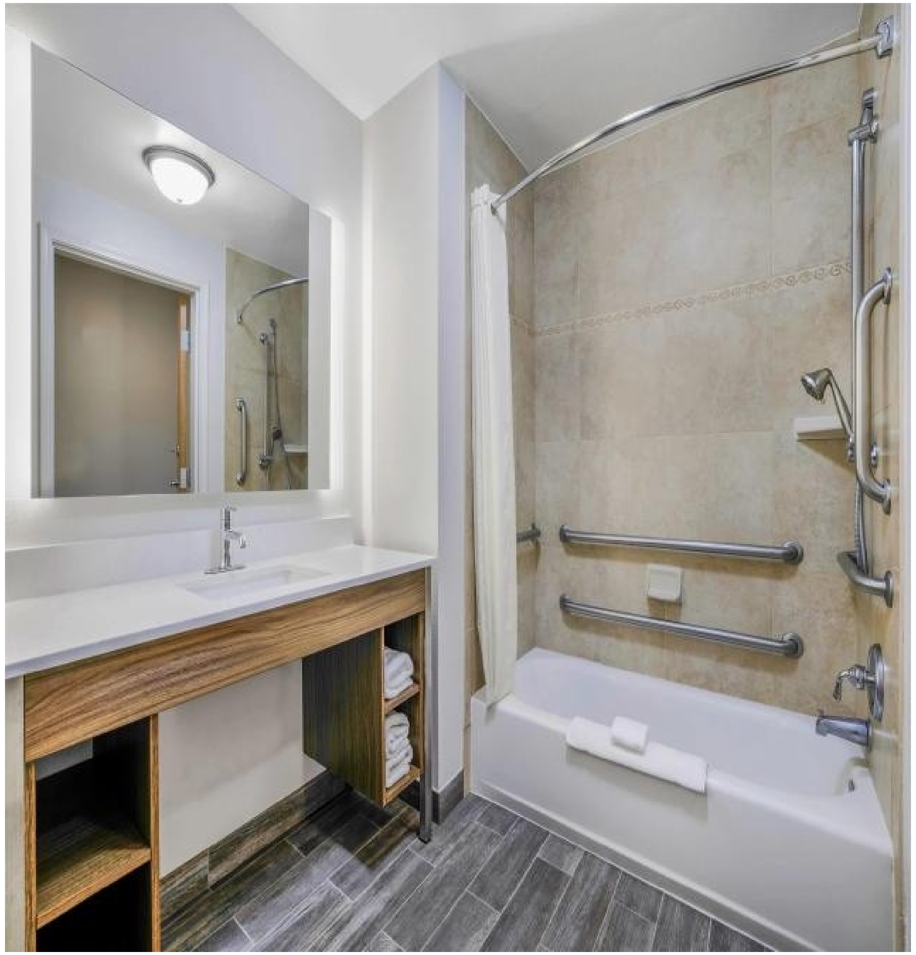
Key Finding

- **Baseline prompts** with controlled **confidence threshold** selection per amenity produced better results
- More description introduced noise, not clarity.

Cropping Strategy

Before → After

- Focuses on the object’s surrounding
- Resulted in better text generation framing
- 35% preserved context without introducing background noise



Source: trivago internal accommodation image dataset. https://www.trivago.com/

Question Generation · Model Exploration Journey

Each stage revealed a limitation that sharpened what the task actually required

Stage 1 · Text Only

Flan-T5 Base & Large

What it produced

Grammatically correct, well-formed questions

Why it failed

Generic and repetitive across all images

No visual reasoning · Detection summary not enough

Need visual grounding →

Stage 2 · Vision + Language

BLIP-2

What it produced

Visually grounded outputs anchored to image content

Why it failed

Captioning-oriented — described what is visible, not what matters

Visual grounding ✓ · Intent-aware reasoning ✗

Need intent-aware reasoning →

Stage 3 · Instruction-Tuned

Gemini 2.5 Flash · 2.5 Pro · 3.0 Pro

What it produced

Decision-relevant, amenity-focused questions from multiple traveller perspectives

Why it worked

Instruction-tuned · follows task directives · reasons about user intent

Visual grounding ✓ · Intent-aware reasoning ✓

Core Insight

The task required not visual perception or language fluency, but intent-aware reasoning grounded in visual evidence.

Why Gemini 3.0 Pro · Model Selection Rationale

Semantic Similarity · BERTScore

Amenity	Flash	Pro	3.0 Pro
Bathtub	0.705	0.708	**0.719**
TV	0.686	0.678	**0.697**
Kettle	0.686	0.696	**0.698**
Hairdryer	**0.717**	0.684	0.694
Mirror	0.704	0.707	**0.728**
Overall	**0.700**	**0.695**	**0.707**

Reference: 25 human-authored questions · 5 per amenity

Behavioral Observations

Gemini 2.5 Flash

Fast and efficient — but tendency toward surface-level reformulations under complex task conditions

Speed ✓ · Depth ✗

Gemini 2.5 Pro

Improved specificity — but collapsed distinctions when additional perspective constraints were introduced

Specificity ✓ · Stability ✗

Gemini 3.0 Pro ✓ Selected

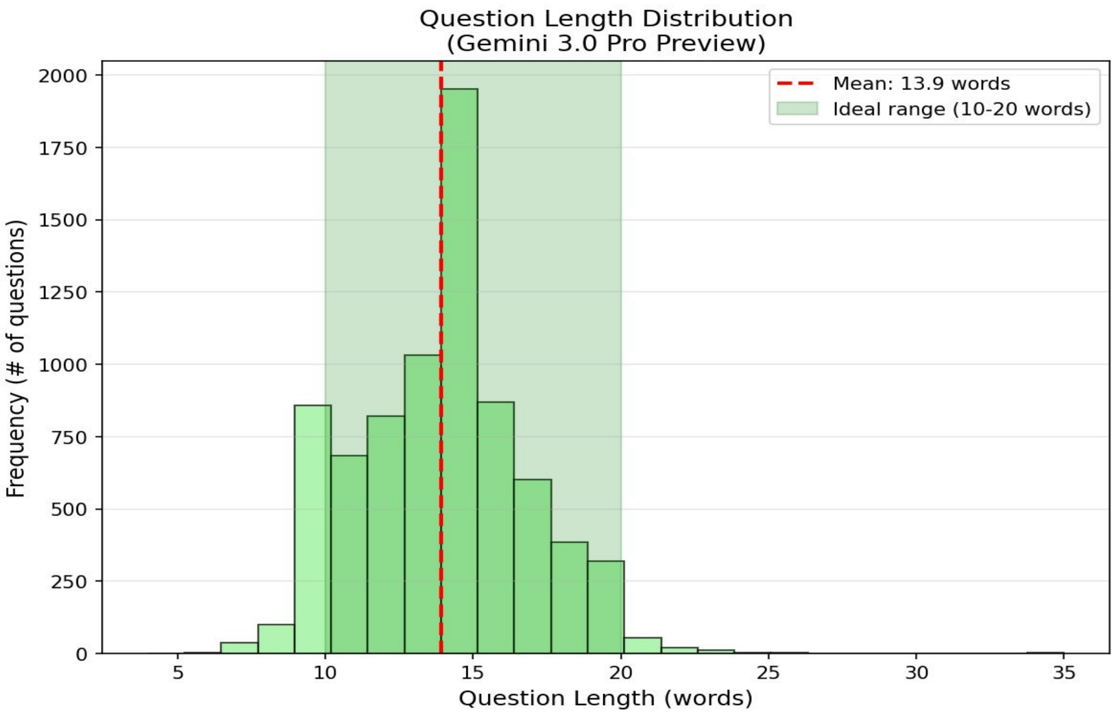
Maintained differentiated, intent-aware outputs across all perspectives and amenity types consistently

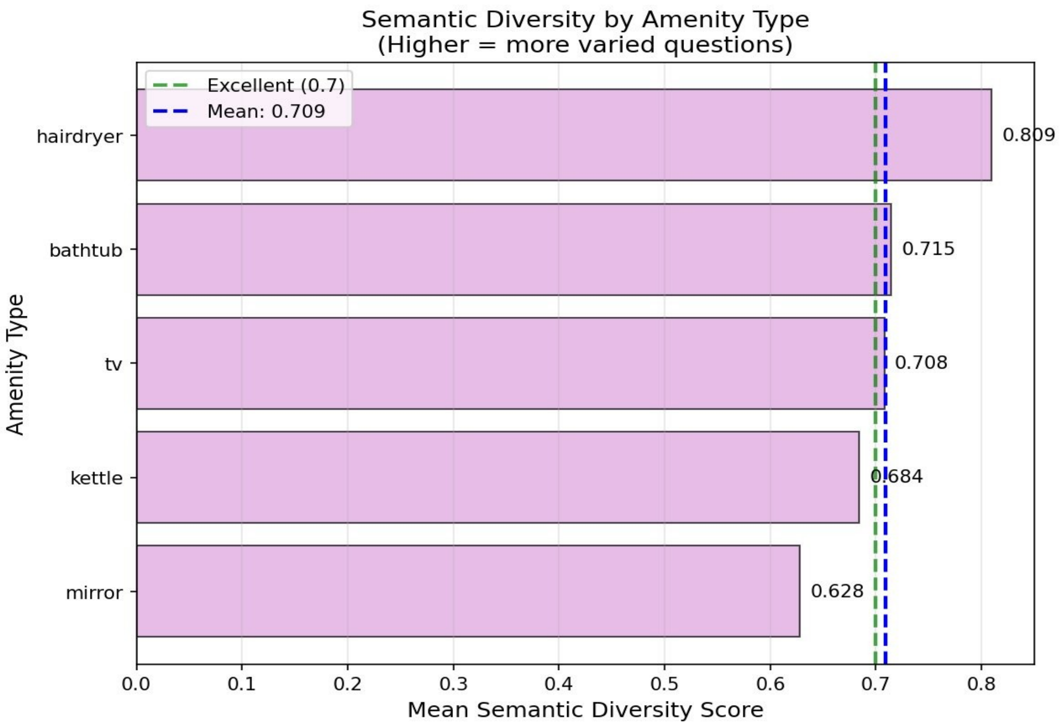
Specificity ✓ · Stability ✓ · Consistency ✓

Selection Rationale

Marginal score differences were consistent, but behavioral stability under task complexity was the decisive factor.

Question Quality Metrics





93% of all questions fell within the 10–20 word range

Human Evaluation · Study Design

What Participants Saw

How the Study Worked

37 Participants

15 images evaluated

3 images per amenity across 5 amenity types

Usefulness Rating

1 = Not helpful at all · 5 = Extremely helpful

2,775 total ratings

EXAMPLE:

1. **Example Question: "Does this bathtub have jets or whirlpool features?**

→ Rate 5 if luxury features matter to you
→ Rate 1 if you never use bathtubs

1	2	3	4	5
☆	☆	☆	☆	☆

Actual Google Form shown to participants · image · generated question · 1 to 5 star rating

Source: trivago internal accommodation image dataset. https://www.trivago.com/

Study Results · Overall Findings

3.36 / 5

Overall Mean Usefulness

Across all 5 amenities and 2,775 ratings

49.6%

Questions Rated 4 or above

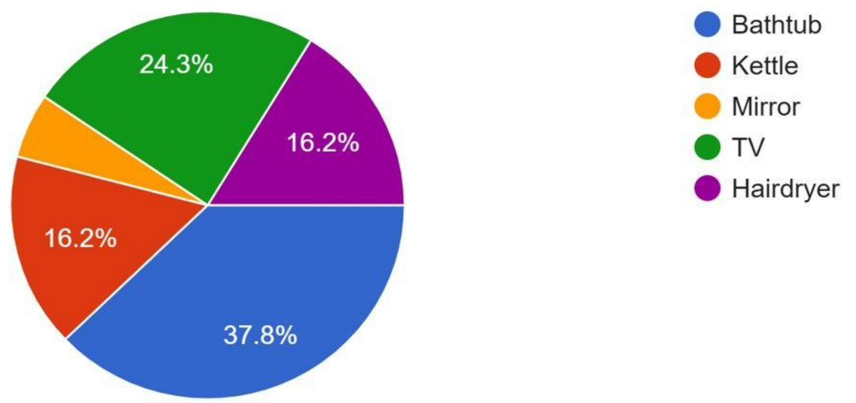
Nearly half rated very or extremely helpful

3.61 → 3.23

Bathtub to Mirror

Highest to lowest mean rating across amenities

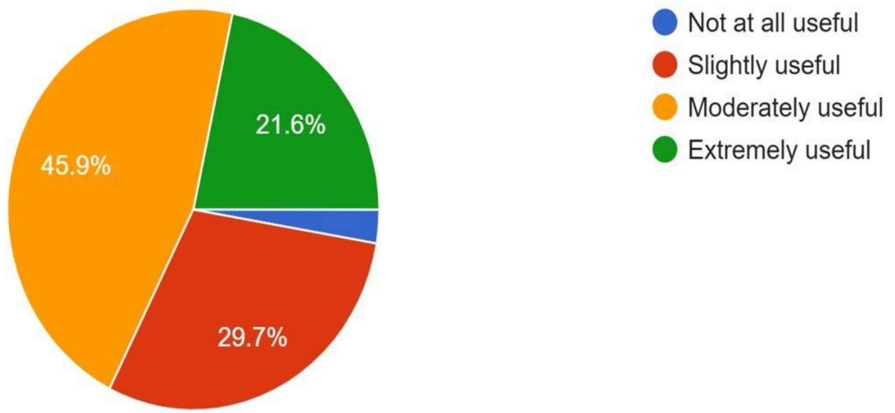
Most Helpful Amenity · Participant Choice



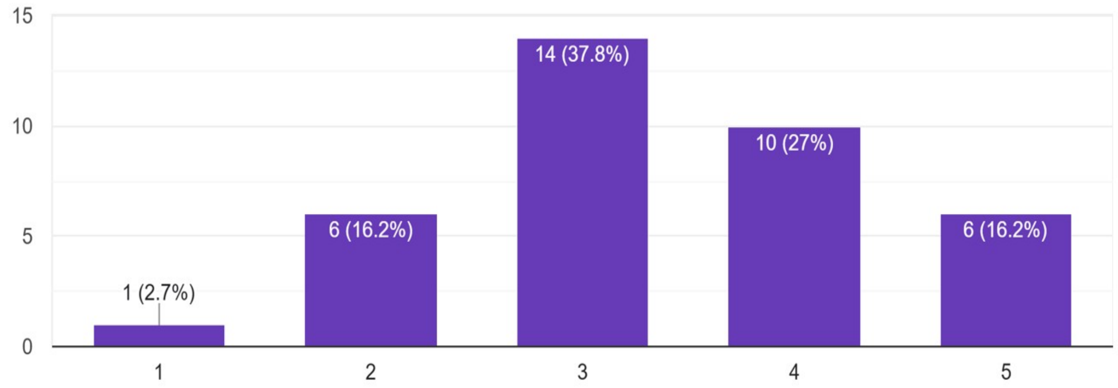
Bathtub questions selected as most helpful by 37.8% of participants

Study Results · Overall Findings

Smart Question Feature Useful on hotel booking website



Compared to typical hotel website descriptions, how would you rate the usefulness of these AI questions?
37 responses



67.5% found the smart question feature moderately or extremely useful

43.2 % found user intended question are more useful than generic captions

Human Evaluation · Bathtub Questions

Mean usefulness · 3.61 / 5

Source: trivago internal accommodation image dataset. https://www.trivago.com/

Bathtub

37 participants · 2,775 ratings

Study images above

Highest Rated

3.86 — Is there a separate walk-in shower in addition to the soaking tub?

3.76 — Does the tub appear deep enough for a full-body soak?

Lowest Rated

2.76 — Do the bathroom tiles and fixtures appear modern or somewhat dated?

2.89 — Are there grab bars installed on the wall for safety and accessibility?

Answering the Research Questions

RQ	Question	Answer
RQ1 *Object Detection*	How can Vision Transformer-based models reliably detect hotel amenities in accommodation images?	Answered • **Baseline prompts** with controlled **confidence threshold** selection per amenity produced better results. • Cropping at 35% padding preserved spatial context for downstream generation.
RQ2 *Question Generation*	How do different vision-language models compare in generating decision-supporting questions from accommodation images?	Answered • Selected **Gemini 3.0 Pro** with highest BERTScore **0.707**, best behavioral stability under task complexity. • Mean semantic diversity **0.709**.
RQ3 *User Validation*	To what extent do generated questions align with real user expectations during accommodation search?	Answered • Overall mean usefulness **3.36 / 5.** • **49.6%** of questions rated 4 or above. • **43.2%** rated AI questions more useful than typical hotel descriptions.

Limitations & Future Work

Limitations - Future Work

User Study Sample

37 participants limits generalizability, Could be properly tested with a bigger user base.

Five amenities only

Can extend to include more amenities

User profiling

User Profile Specific Questions can be achieved in future scope

English language only

Question generation and evaluation conducted entirely in English - **multilingual extension** is possible

Question answering layer

Add retrieval layer to answer generated questions from existing database - completing the decision-support loop

Conclusion

01

Visual question generation from accommodation images is feasible

02

Generated questions provide measurable decision support value

03

User evaluation showcased approval and acceptance towards the approach

Thank You

Q & A

5 Perspectives for prompt config for Gemini

The 5 Areas the Perspectives Cover

#	Focus Area	What it asks the model to consider
1	Functionality & Practical Usability	Does it work? How do you use it? Is it fit for purpose?
2	Size, Capacity & Spatial Constraints	Is it big enough? Where is it placed? Does it fit the space?
3	Quality Indicators & Maintenance Cues	Does it look clean? Is it in good condition? Does it look worn?
4	Luxury or Premium Features	Does it have premium attributes? Is it above standard?
5	Notable or Distinguishing Features	Does it have anything unusual or standout visible in the image?

Human Evaluation · Kettle Questions

Mean usefulness · 3.39 / 5

Source: trivago internal accommodation image dataset. https://www.trivago.com/

Kettle

37 participants · 2,775 ratings

Study images above

Highest Rated

4.08 — Does the appliance look modern and in good, clean condition?

3.81 — Are there accessible power outlets immediately available at the beverage station?

Lowest Rated

3.05 — Is there a premium coffee machine available in addition to the standard electric kettle?

3.11 — Are there matching canisters for tea, coffee, or sugar visible next to the kettle?

Human Evaluation · Mirror Questions

Mean usefulness · 3.23 / 5

Source: trivago internal accommodation image dataset. https://www.trivago.com/

Mirror

37 participants · 2,775 ratings

Study images above

Highest Rated

3.49 — Is the overhead lighting bright enough for detailed tasks like applying makeup or shaving?

3.49 — Is the mirror wide enough for two people to get ready at the same time?

Lowest Rated

2.92 — Since this appears to be a standard flat mirror, is a separate magnifying mirror provided for close-up needs?

3.03 — Does the mirror fog up quickly when the nearby shower (visible in the reflection) is in use?

Human Evaluation · TV Questions

Mean usefulness · 3.36 / 5

Source: trivago internal accommodation image dataset. https://www.trivago.com/

TV

37 participants · 2,775 ratings

Study images above

Highest Rated

3.92 — Is this a Smart TV that allows guests to log into their own streaming accounts?

3.76 — Are the HDMI or USB ports easily accessible for connecting personal devices like a laptop?

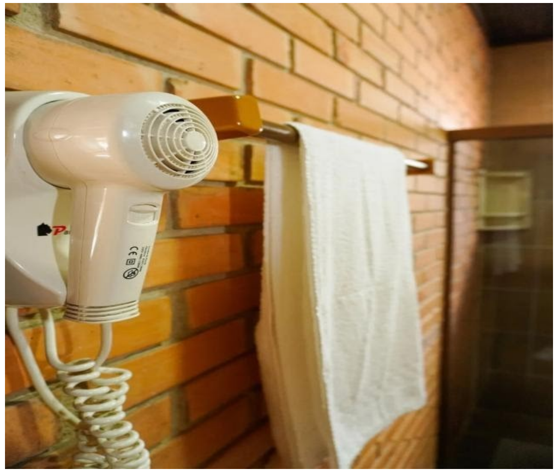
Lowest Rated

2.19 — Is this a Frame style TV that displays artwork when not in use?

2.22 — Is the white border a digital display effect or a physical part of the frame?

Human Evaluation · Hairdryer Questions

Mean usefulness · 3.32 / 5



Source: trivago internal accommodation image dataset. https://www.trivago.com/

Hairdryer

37 participants · 2,775 ratings

Study images above

Highest Rated

3.73 Does the switch on the handle offer multiple heat or speed settings?

3.62 Is the hairdryer permanently wall-mounted, limiting where I can dry my hair?

Lowest Rated

2.84 Does the dryer have a folding handle to save space on the bathroom counter?

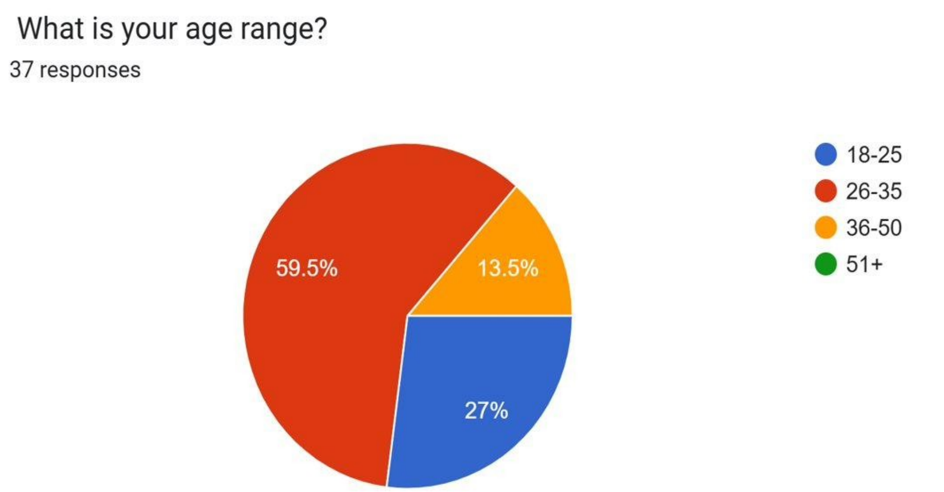
3.35 Does the 1875 wattage rating mean it is powerful enough to dry thick hair quickly?

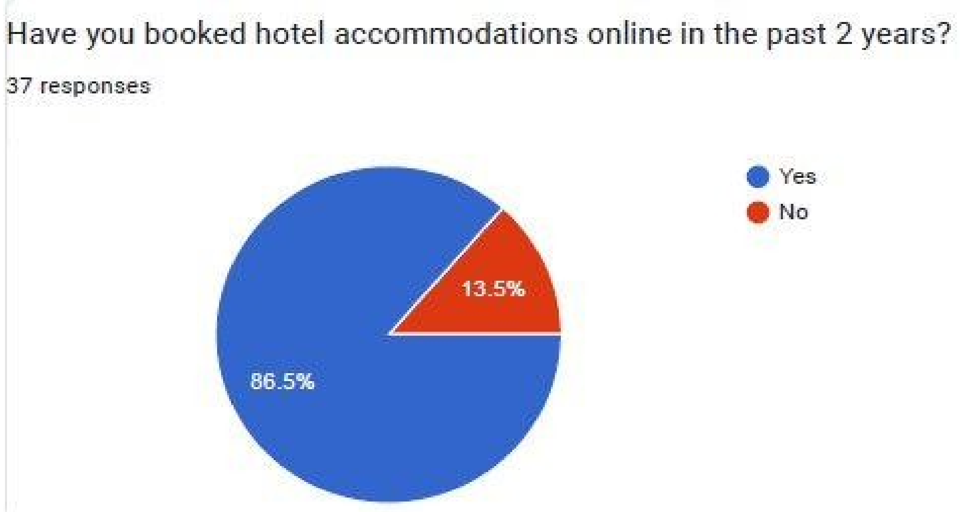
OWL-ViT Per Amenity optimal config

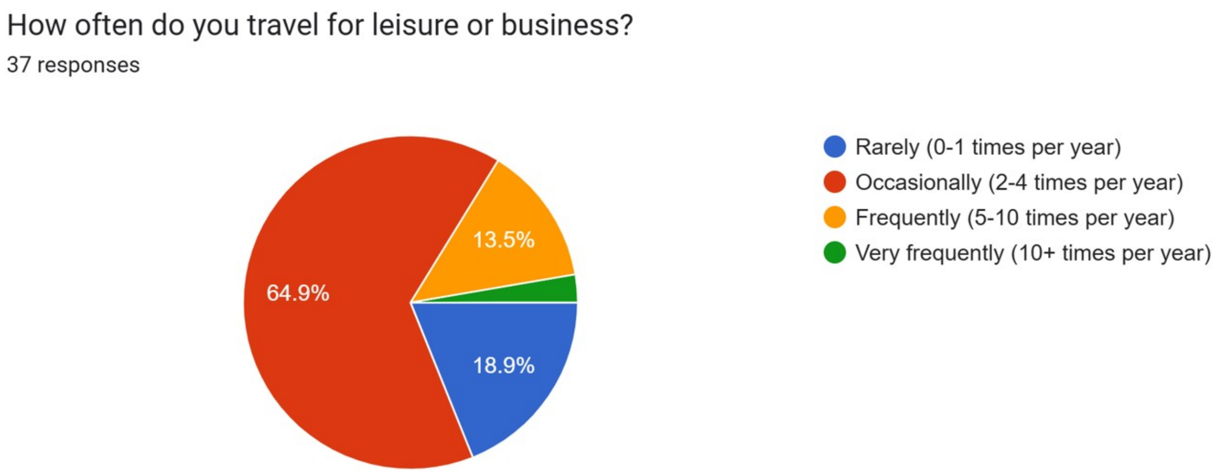
Optimal Configuration Per Amenity

Amenity	Best Prompt	Threshold	F1	Precision	Recall	Mean IoU
Bathtub	Baseline	0.05	0.790	0.721	0.873	0.641
Kettle	Electric	0.05	0.699	0.738	0.665	0.627
Hairdryer	Variants	0.05	0.808	0.747	0.881	0.648
Mirror	Baseline	0.15	0.731	0.715	0.749	0.633
TV	Baseline	0.05	0.836	0.876	0.799	0.750

Human Evaluation · Participant Profile







Research Questions

RQ1

Object Detection

How can Vision Transformer-based models reliably detect hotel amenities in accommodation images?

Prompt design configuration · Confidence thresholds · Context Preservation

RQ2

Question Generation

How do different vision-language models compare in generating decision-supporting questions from accommodation images?

Model comparison · Prompt engineering · Diversity & Grounding

RQ3

User Validation

To what extent do generated questions align with real user expectations during accommodation search?

Perceived usefulness · Relevance